

# Task 3: Artificial Intelligence Chatbot

Complete Java source code — Swing GUI, NLP and rule-based responses

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import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.util.*;

public class AIChatbot extends JFrame {

    private JTextArea chatArea;
    private JTextField inputField;
    private JButton sendButton;
    private JButton clearButton;

    public AIChatbot() {

        setTitle("AI Chatbot - Task 3");
        setSize(600, 500);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        JPanel mainPanel = new JPanel(new BorderLayout(10, 10));
        mainPanel.setBorder(
            BorderFactory.createEmptyBorder(10, 10, 10, 10)
        );

        chatArea = new JTextArea();
        chatArea.setEditable(false);
        chatArea.setFont(new Font("Arial", Font.PLAIN, 15));
        chatArea.setLineWrap(true);
        chatArea.setWrapStyleWord(true);

        JScrollPane scrollPane = new JScrollPane(chatArea);

        inputField = new JTextField();
        inputField.setFont(new Font("Arial", Font.PLAIN, 15));

        sendButton = new JButton("Send");
        clearButton = new JButton("Clear");

        JPanel bottomPanel =
            new JPanel(new BorderLayout(5, 5));

        bottomPanel.add(inputField, BorderLayout.CENTER);

        JPanel buttonPanel = new JPanel(new FlowLayout());
        buttonPanel.add(sendButton);
        buttonPanel.add(clearButton);

        bottomPanel.add(buttonPanel, BorderLayout.EAST);

        mainPanel.add(scrollPane, BorderLayout.CENTER);
        mainPanel.add(bottomPanel, BorderLayout.SOUTH);

        add(mainPanel);

        chatArea.append(
            "■ AI Chatbot: Hello! Welcome to my AI Chatbot.\n"
        );
        chatArea.append(
            "■ AI Chatbot: Ask me something! " +
            "Type 'help' to see what I can answer.\n\n"
        );

        sendButton.addActionListener(e -> sendMessage());
        inputField.addActionListener(e -> sendMessage());

        clearButton.addActionListener(e -> {
            chatArea.setText("");
            chatArea.append(
                "■ AI Chatbot: Chat cleared. " +
                "How can I help you?\n\n"
            );
        });
    }

    private void sendMessage() {

        String userMessage =
            inputField.getText().trim();

        if (userMessage.isEmpty()) {
            return;
        }
    }
}
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chatArea.append(
    "■ You: " + userMessage + "\n"
);

String response =
    getResponse(userMessage);

chatArea.append(
    "■ AI Chatbot: " + response + "\n\n"
);

inputField.setText("");
}

private String getResponse(String message) {

    // NLP STEP 1: Text normalization
    String text =
        message.toLowerCase().trim();

    // NLP STEP 2: Remove special characters
    text =
        text.replaceAll("[^a-zA-Z0-9 ]", "");

    // NLP STEP 3: Tokenization
    String[] words =
        text.split("\\s+");

    // RULE 1: Greeting
    if (containsAny(
        words,
        "hello", "hi", "hey", "hii", "namaste")) {

        return "Hello! ■ How can I help you today?";
    }

    // RULE 2: Name
    if (text.contains("your name") ||
        text.contains("who are you")) {

        return "My name is AI Chatbot. " +
            "I am a Java-based rule-driven chatbot.";
    }

    // RULE 3: How are you
    if (text.contains("how are you")) {

        return "I am doing great! ■ " +
            "Thanks for asking.";
    }

    // RULE 4: Java
    if (containsAny(
        words, "java", "programming", "coding")) {

        return "Java is a powerful object-oriented " +
            "programming language used for web, " +
            "desktop, mobile and enterprise applications.";
    }

    // RULE 5: NLP
    if (containsAny(
        words, "nlp", "language", "processing")) {

        return "NLP stands for Natural Language Processing. " +
            "It helps computers understand and process " +
            "human language.";
    }

    // RULE 6: AI
    if (containsAny(
        words, "ai", "artificial", "intelligence")) {

        return "Artificial Intelligence allows computers " +
            "to perform tasks that normally require " +
            "human intelligence.";
    }

    // RULE 7: Machine Learning
    if (containsAny(
        words, "machine", "learning", "ml")) {

        return "Machine Learning is a branch of AI where " +
            "computers learn patterns from data and use " +
            "them to make predictions or decisions.";
    }

    // RULE 8: Full Stack
    if (containsAny(
        words,
        "fullstack", "full", "stack", "developer")) {

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        return "A Full Stack Developer works with both " +
               "frontend and backend technologies such as " +
               "HTML, CSS, JavaScript, Java, databases and APIs.";
    }

    // RULE 9: College
    if (containsAny(
        words, "college", "school", "study", "student")) {

        return "Keep learning and practicing every day! " +
               "■ Consistency is the key to becoming " +
               "a good developer.";
    }

    // RULE 10: Internship
    if (containsAny(
        words, "internship", "job", "career")) {

        return "For a software development career, build " +
               "projects, practice coding, create a GitHub " +
               "portfolio and apply for internships.";
    }

    // RULE 11: GitHub
    if (containsAny(
        words, "github", "repository", "repo")) {

        return "GitHub is a platform used to store, " +
               "manage and share source code using Git.";
    }

    // RULE 12: Thanks
    if (containsAny(
        words, "thanks", "thank", "thx")) {

        return "You're welcome! ■";
    }

    // RULE 13: Help
    if (containsAny(
        words, "help", "commands", "options")) {

        return "You can ask me about Java, AI, NLP, " +
               "Machine Learning, Full Stack Development, " +
               "GitHub, internships and programming.";
    }

    // RULE 14: Bye
    if (containsAny(
        words, "bye", "goodbye", "exit", "quit")) {

        return "Goodbye! ■ Keep coding and keep learning!";
    }

    // DEFAULT RULE
    return "Sorry, I don't understand that yet. ■ " +
           "Try asking about Java, AI, NLP, Machine Learning, " +
           "GitHub or Full Stack Development.";
}

private boolean containsAny(
    String[] words,
    String... keywords) {

    for (String word : words) {

        for (String keyword : keywords) {

            if (word.equals(keyword)) {
                return true;
            }
        }
    }

    return false;
}

public static void main(String[] args) {

    SwingUtilities.invokeLater(() -> {

        AIChatbot chatbot =
            new AIChatbot();

        chatbot.setVisible(true);
    });
}
}

```